光之殇

Lost in the Glare



昆虫绕着灯飞，真只是因为喜欢光吗？

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RECALL 回想巩固

英文原文

A moth skims over an ultraviolet lamp and flips upside down, belly to the sky. In Costa Rica's misty Monteverde cloud-forest, stereo cameras caught such acrobatics for a 2024 Imperial College study that unravelled an age-old puzzle, the helpless swirl of insects round porch lights.

Folk wisdom blames the circling on attraction. In fact, insects instinctively keep their backs to whatever shines brightest, normally the sky, to stay level. A lamp, a counterfeit sky, hijacks this built-in habit, leaving insects eddying in its halo or inverted above it. In 477 field clips, precious few darted straight at the light.

Many fireflies flirt with their lanterns. A male blinks an overture, a female winks back, and bright nights smother the banter. At Tufts University, suitors under artificial light mustered about five flash patterns a minute instead of ten. Females reciprocated about half as often under red or blue light, scarcely at all under white, and never under amber.

The harm spills into the ecosystem. Along 55 Thames Valley roadsides, lamp-lit hedges held 47% fewer moth caterpillars than dark ones, with harsh white LEDs worse than mellower sodium lamps. Street lamps rigged up in seven of 14 Swiss meadows cut nocturnal pollinator visits by 62%. The plants set fewer fruit, and daytime pollinators proved poor understudies.

Over 80% of humanity lives beneath a luminous veil, which hides the pale ribbon of the Milky Way from more than a third. Citizen scientists' star counts from 2011 to 2022 show the sky brightening about 9.6% a year. At that pace, a child born where 250 stars twinkle would see about 100 by their eighteenth birthday.

中文译文

一只飞蛾掠过紫外灯，突然翻转，肚皮朝向天空。2024年伦敦帝国理工学院的一项研究，在哥斯达黎加蒙特韦尔德雾气弥漫的云雾林中，用立体摄像机捕捉到这样的飞行动作，解开了一个古老谜团：昆虫为什么会身不由己地绕着门廊灯盘旋。

人们通常认为，昆虫绕灯飞是受到灯光吸引。实际上，为保持身体平稳，昆虫会本能地让背部朝向最亮的方向，通常就是天空。人造灯冒充了天空，扰乱这种天生习性，使昆虫在光晕中打转，或在灯上方翻倒。477段野外录像中，真正径直冲向光源的昆虫寥寥无几。

许多萤火虫用发光来求偶：雄虫先闪光，雌虫再回应，太亮的夜晚却会淹没这番交流。在塔夫茨大学的实验中，人造光下的雄虫每分钟只能发出约五组闪光，而原本约有十组。实验中的雌虫在红光或蓝光下回应的次数约减半，在白光下几乎没有回应，在琥珀色光下则完全没有回应。

影响还延伸到生态系统。对泰晤士河谷55处路边的调查发现，有灯照明的树篱里，蛾类毛虫比无灯处少47%；刺眼的白色LED处，情况比柔和的钠灯处更严重。在瑞士14片草甸中的7片临时架起路灯后，夜间传粉者的造访减少62%。植物结的果实也更少，白天的传粉者未能补足夜间的损失。

全球超过80%的人口生活在发光的夜空帷幕下，超过三分之一的人看不到银河那条淡淡的光带。参与观星的公众在2011年至2022年间记录的星数表明，夜空每年约变亮9.6%。按这一速度，一个孩子出生时若能在当地看到250颗星，到18岁时便大约只能看到100颗。

树篱毛虫少47%来自路边调查；草甸的62%来自架灯实验，不应把两种证据都写成同一种因果证明。

核心词汇

先记这22项。连着下方的原文短语一起回想。

skim over 动词短语
掠过
A moth skims over an ultraviolet lamp

unravel v.
解开（谜团）
unravelled an age-old puzzle

age-old adj.
由来已久的
an age-old puzzle

folk wisdom n.
民间流传的观念
Folk wisdom blames the circling

attraction n.
吸引
blames the circling on attraction

instinctively adv.
本能地
insects instinctively keep their backs

counterfeit adj.
假冒的
a counterfeit sky

hijack v.
劫持；此处指扰乱
hijacks this built-in habit

built-in adj.
与生俱来的
this built-in habit

inverted adj.
翻转倒置的
inverted above it

precious few 数量短语
少之又少
precious few darted straight

reciprocate v.
回应
Females reciprocated

ecosystem n.
生态系统
The harm spills into the ecosystem

caterpillar n.
毛虫；蛾蝶幼虫
47% fewer moth caterpillars

harsh adj.
刺眼的；强烈的
harsh white LEDs

sodium lamp n.
钠灯
mellower sodium lamps

rig up 动词短语
临时架设
Street lamps rigged up

nocturnal adj.
夜间的
nocturnal pollinator visits

understudy n.
替补
proved poor understudies

luminous adj.
发光的
a luminous veil

citizen n.
公民；此处指公众参与者
Citizen scientists' star counts

pace n.
速度
At that pace

文中词汇与搭配

补充词汇共32项。保留每个表达在本文中的意思。

belly n.
腹部

misty adj.
雾气弥漫的

cloud forest n.
云雾林

stereo adj.
立体成像的

acrobatics n.
杂技般的动作

helpless adj.
身不由己的

swirl n.
盘旋；打转

porch n.
门廊

eddy v.
打旋

halo n.
光晕

clip n.
视频片段

dart v.
猛冲；疾驰

flirt v.
求偶；调情

lantern n.
灯笼；此处喻发光器官

blink v.
闪烁

overture n.
序曲；此处喻求偶信号

wink v.
眨眼；此处喻闪光回应

smother v.
压制；淹没

banter n.
打趣；此处喻求偶交流

suitor n.
求偶者

muster v.
勉强发出（闪光）

amber n.
琥珀色

spill into 动词短语
蔓延到

roadside n.
路边

lamp-lit adj.
有灯照明的

hedge n.
树篱

mellow adj.
较柔和的

meadow n.
草甸

humanity n.
全人类

veil n.
帷幕

ribbon n.
带状物；光带

twinkle v.
闪烁

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 In fact, insects _____ keep their backs to whatever shines brightest, normally the sky, to stay level.

02 A lamp, a _____ sky, hijacks this built-in habit, leaving insects eddying in its halo or inverted above it.

03 Females _____ about half as often under red or blue light, scarcely at all under white, and never under amber.

04 Street lamps rigged up in seven of 14 Swiss meadows cut _____ pollinator visits by 62%.

文章主线

- 迷向 昆虫本能地背朝最亮处以保持平稳；2024年研究发现，灯冒充天空，扰乱这种习性，使昆虫绕灯打转或翻身。
- 求偶 萤火虫靠闪光求偶；实验中人造光使雄虫闪光约减半，雌虫在红蓝光下回应减半，白光下极少、琥珀光下全无。
- 幼虫 英国55处路边调查中，有灯树篱的蛾类毛虫比暗处少47%；白色LED的影响重于钠灯。
- 传粉 瑞士草甸架灯实验使夜间传粉访问减少62%、植物结实减少；白天传粉未能补足夜间损失。
- 夜空 全球逾八成人口受光污染影响，逾三分之一看不见银河；2011—2022年公众观星记录显示，夜空年均增亮约9.6%。

参考答案　1 instinctively 2 counterfeit 3 reciprocated 4 nocturnal